NAU NORTHERN ARIZONA UNIVERSITY
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ME486C

PHS Sustainability Capstone – Project Management

I. Introduction

The purpose of the project management for ME486C assignment is to outline the plans and necessary next steps for the PHS sustainability capstone project. This includes a reflection on the spring semester, highlighting key successes, lessons learned, and overall outcomes achieved by the team. Additionally, the management plan presents a structured approach for the upcoming semester, including a Gantt chart, purchasing strategy, and manufacturing plan. Together, these elements provide a clear framework to help the team stay organized and prepared as they transition into ME486C.

II. Reflection

Successes

Listed below are successes and achievements from the PHS sustainability capstone project (in terms of project management and team communication):

- All assignments turned in on time with all individual and team components completed
- Consistent communication with PHS group chat, teams posting, and individual messaging
- Weekly team meetings with goals/updates for the week
- Early communication of availability from every team member
- Team stayed on track with majority of project goals for the semester
- Biweekly client meetings with goals/updates
- Collaboration from all team members in all aspects of the project
- Clear communication with project manager for any potential issue for any team member
- Quick resolution for any communication issues in the team

Room for Improvements

Listed below are areas for improvement in the PHS sustainability capstone project (in terms of project management and team communication):

- More communication from all team members when additional assistance is needed
- Larger commitment to stay on top of all project goals
- Clearer communication with Prof. Carson Pete from all team members
- More check-ins from project manager with each team member
- Clearer small project goals (this improved throughout the semester)
- More team building events through prototyping, community service, and club events
- Less individual responsibility when it comes to team assignments (all of us are responsible for the success of the project and it shouldn’t fall back on one team member)

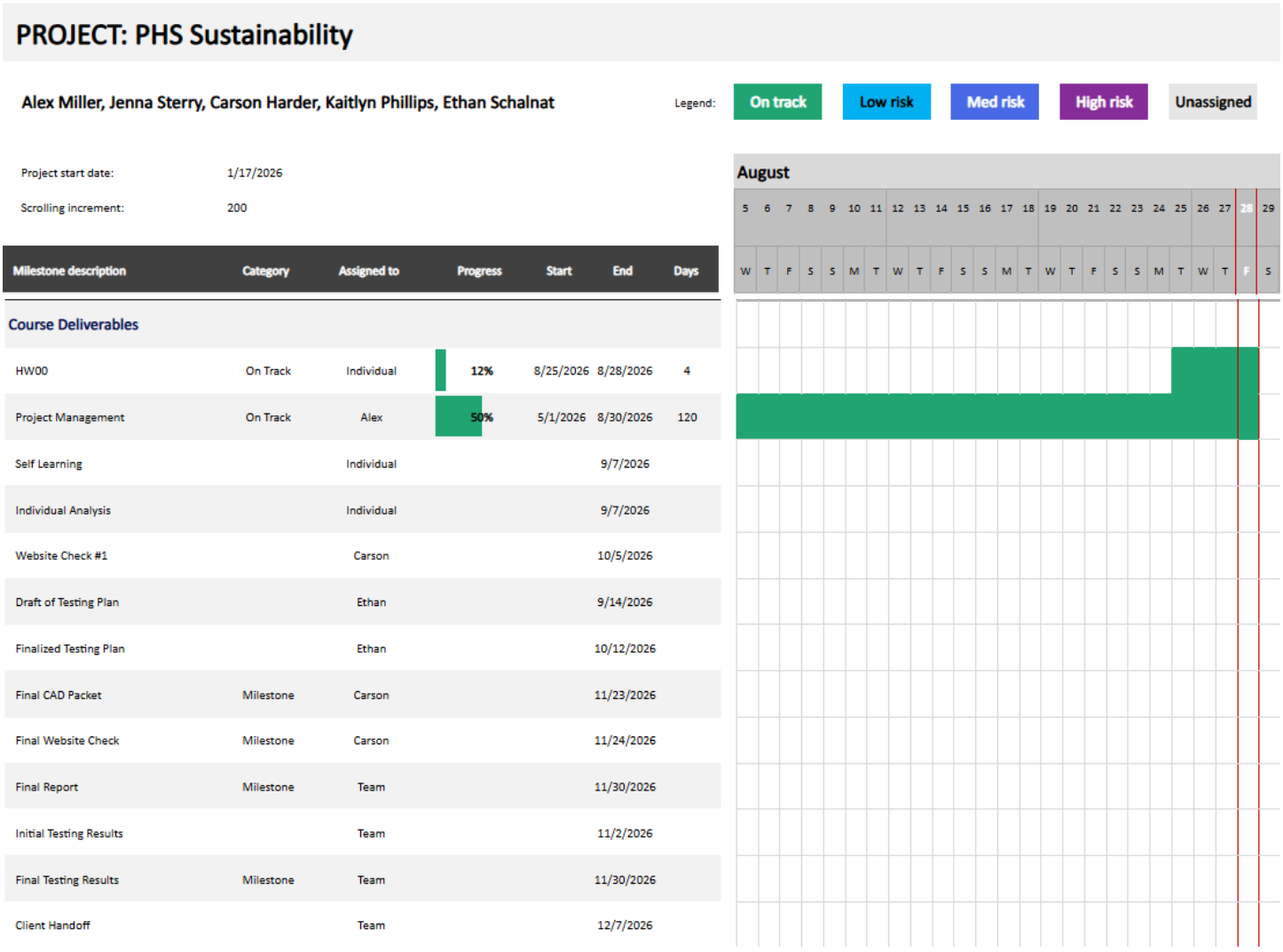
Action Items

1) Check in 3-4 days before assignment submission with all team members to give needed assistance
2) Define goals biweekly with tasks to achieve respective goals
3) Biweekly check in with prof. Carson Pete as well as consistent teams messaging
4) Biweekly check in with each team member from the project manager (online or in person)
5) Planned events through energy club for community service (terra birds, campus clean up, etc.) as well as after work dinners

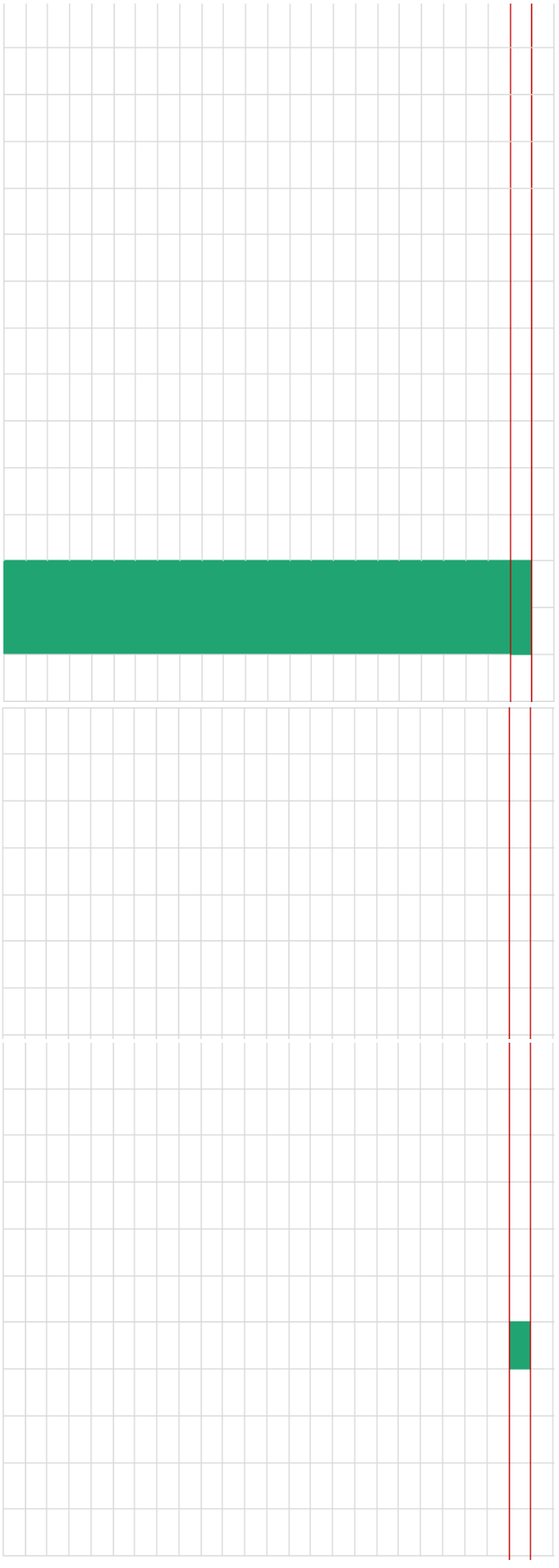
Remaining Design Efforts

- Circuit diagram for the PHS greenhouse
- Improved frame stand/design with new budget improvement
- Solar panel diagram for the PHS greenhouse roof (complete with prof. Carson Pete)

III. Gantt Chart



Build Deliverables						
33+% Build	Milestone	Team			9/14/2026	
Peer eval 1		Individual			9/14/2026	
67+% Build	Milestone	Team			10/5/2026	
Peer eval 2		Individual			10/5/2026	
100% Build	Milestone	Team			10/26/2026	
Peer eval 3		Individual			10/26/2026	
Peer eval 4		Individual			12/7/2026	
Client Deliverables						
Energy Audit Report	Goal	Team	100%	2/20/2026	4/27/2026	57
System Capacity and Performance Analysis	Goal	Team	100%	3/16/2026	5/3/2026	33
Improvement/Expansion Reccommendations	Milestone	Kaitlyn and Alex	100%	3/16/2026	5/3/2026	37
Human-Powered Bicycle Generator	On Track	Ethan and Carson	40%	2/3/2026	12/7/2026	207
Education Curriculum Materials	On Track	Jenna and Alex	25%	2/1/2026	11/24/2026	209
Client Deliverable Completeion	Milestone	Team			12/7/2026	
Ugrad/Efest deliverables						
Ugrad/Efest regristration		Alex			9/28/2026	
Draft of Poster		Team			10/5/2026	
Final Poster	Milestone	Team			10/26/2026	
Final Presentation	Milestone	Team			10/26/2026	
Efest Practice (Demo and Presentation)		Team			11/9/2026	
Efest	Milestone	Team			11/16/2026	
Education Curriculum Deliverables						
Lesson Plan draft	On Track	Jenna and Alex	100%	2/1/2026	2/16/2026	16
February Lesson	On Track	Jenna and Alex	100%	2/20/2026	2/26/2026	7
March Lesson	On Track	Jenna and Kaitlyn	100%	3/17/2026	3/24/2026	8
KidWind participation	On Track	Team	100%	4/1/2026	4/2/2026	2
April/May Lesson	On Track	Jenna and Alex	100%	4/13/2026	4/21/2026	9
August/September Lesson	On Track	TBD		8/28/2026		1
October Lesson		TBD				
November Lesson		TBD				
Cummulative Education Report	Milestone	Team			11/24/2026	
Community Impact Presentation		Jenna			11/24/2026	



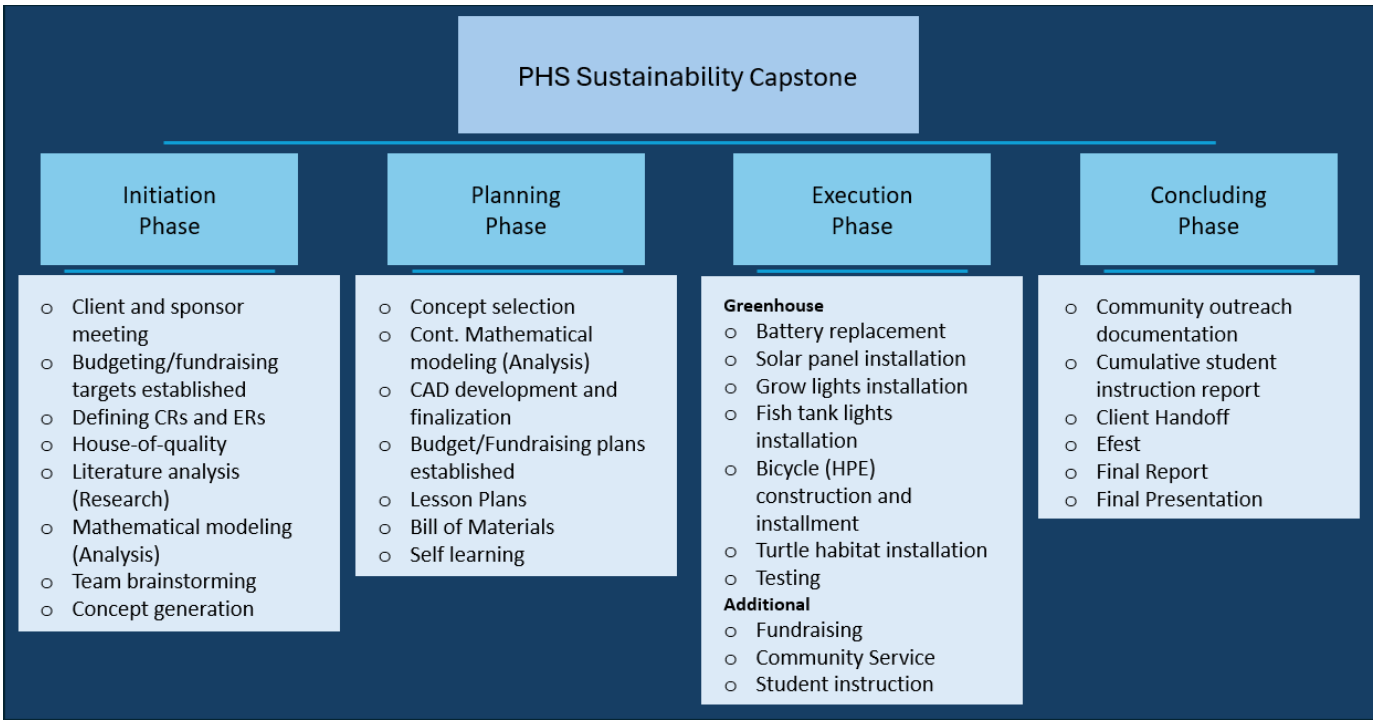
Note for Gantt chart: these dates are adjusted to the tentative schedule for ME486C. With Efest being moved to the week of November 16th, dates for other deliverables have been moved accordingly. All dates are subject to change.

Expanded Gantt chart for week 1 of ME486C until 1st Hardware Status Update

Milestone description	Category	Assigned to	Progress	Start	End	Days
Build Milestones						
Tortoise habtitat enclosure	On Track	Jenna and Alex			9/14/2026	
Improvements/reccomendations	On Track	Kaitlyn and Alex	100%	8/24/2026	8/28/2026	5
Battery storage installation	On Track	Team and Pete			9/14/2026	
Bike frame purchased	On Track	Kaitlyn	20%	8/27/2026	9/7/2026	2
Switches and Outlet Installation		Team and Pete			9/14/2026	
Grow light installation		Team			9/14/2026	
33+% Build	Milestone	Team		8/24/2026	9/14/2026	5
Peer eval 1		Individual		9/14/2026	9/14/2026	

Note: Dates for the tortoise habitat enclosure, battery, grow lights, and switches are not set because installation dates are being set with client John and professor Pete in a meeting on September 4th.

Work Breakdown Schedule (WBS)



Major Work/Milestones to the First Hardware Status Update

Below is a list of all important milestones that need to be accomplished prior to the first hardware status update.

- Confirm list of improvements/recommendations for the PHS greenhouse
- Have established plans for wiring of the solar panels and control system
- Install the new battery storage
- Install switches and outlets
- Manufacture and install the tortoise greenhouse enclosure
- Manufacture and install the grow light pulley system

IV. Top Level Finances

Budget and Fundraising

Reviewing and tracking the budget as the project approaches completion is essential to ensure that both the University and the client remain within or under the agreed funding amount. Throughout the project, the team has monitored expenditures and anticipated costs to ensure that sufficient funding remains for final manufacturing, purchasing, testing, and implementation. We would also like to provide final materials to the school to assist with future troubleshooting or part replacement. The team also successfully met the required fundraising goal of $750.00 through two fundraising events: a fundraiser at Chipotle and a plant sale in partnership with the NAU Green Club and NAU Botany Club.

Additionally, the team's NAU provided budget increased by $1,500.00 from the previous semester. The additional funding provides greater flexibility as the project transitions from design into final planning and implementation. One significant change made possible by the increased budget is the decision to purchase a commercially available bicycle frame rather than manufacture a custom frame, as originally planned. Purchasing the frame will streamline the manufacturing process, reduce fabrication time, and allow the team to focus its resources on the other capstone components. This change represents an important budget-driven design decision that can reduce manufacturing risk while supporting completion of the final bicycle generator within the project schedule.

PHS Sustainability Budget

Revenue		Estimated	Actual	
NAU		$2,500.00	$2,500.00	
Fundraising		$750.00	$750.51	Chipotle: $450.51, Plant Sale: $300.00
Ponderosa High School		$4,000.00	$3,784.49	
		$7,250.00	**$7,035.00**	

Expense Type		Budget	Expense	Remaining
Tools		$300.00	$60.14	$239.86
Batteries		$3,500.00	$3,559.19	($59.19)
Solar Panel		$1,100.00	$1,031.95	$68.05
LED Lights		$100.00	$0.00	$100.00
Human Powered Energy Bike		$1,000.00	$18.96	$981.04
Animal Insulation		$500.00	$0.00	$500.00
Curriculum		$500.00	$13.00	$487.00
Electrical		$500.00	$306.21	$193.79
Total Expenses:		**$7,500.00**	**$4,989.45**	**$2,510.55**
			Total Revenue:	$7,035.00
			Total Balance:	**$2,045.55**

Purchasing Plan

Purchasing components is a critical aspect of engineering project execution and closely mirrors real-world industry practices in parts acquisition and supply chain management. For this project, a comprehensive Bill of Materials (BOM) has been developed and expanded to include key procurement-related attributes that support informed decision-making and efficient project planning. This BOM is included as an appendix at the back of the report.

Each item in the BOM includes the following information:

- Cost: Identified the price of each component. Donated items are clearly labeled to distinguish them from purchased components.
- Make/Buy decision: Indicated whether the part will be manufactured in-house or purchased from an external supplier.
- Primary Vendor: Specifies the preferred supplier for each purchased components, selected based on availability, cost, and delivery time.
- Manufacturer: Identifies the original manufacturer of the component, which is important for quality assurance and specification verification.
- Lead Time: Provides the expected delivery time from the primary vendor, which is important for quality assurance and specification verification.

- Part Status: Tracks the current state of each component (i.e. planned for purchase, on order, received, in manufacturing, or assembled). This field is continuously updated throughout the project lifecycle.

For this report, the BOM has been filtered to include only purchased items, allowing for a focused analysis of procurement requirements.

Observations and Planning Needs

Based on the current BOM, several areas require additional planning:

- Some components lack clearly defined vendors or manufacturers
- Lead times are missing or uncertain for critical items
- Several high-priority components have not yet been ordered
- Cost estimates may need validation for accuracy
- Make/Buy decisions are not finalized for all parts, particularly the frame for the bike, whether a manufactured or purchased metal bike frame will be used

Action Items for 486C

1) Finalize vendor selection for all purchased components lacking a primary supplier
2) Determine and document lead times for all critical items
3) Validate and update cost estimates to ensure budget accuracy
4) Confirm manufacturer/buy decisions for all undecided components
5) Initiate purchase orders for long lead-time items immediately
6) Establish a regular update schedule for the parts that do have lead times

V. Manufacturing Plan

The following manufacturing plan provides a structured overview of all internally produced components for the project. This plan identifies key details for each manufactured part, including the party responsible, material requirements, production location, and projected timeline. By consolidating this information into a single reference table, the team can coordinate efforts, anticipate resource needs, and track progress toward production milestones. The table below has been filtered to display manufactured items only and should be reviewed regularly as designs are finalized and schedules evolve.

Part	Manufacturer	Manufacturing Duration (wks)	Expected Finish	Raw Material	Manufacturing Location
Tortoise Habitat Enclosure	Team, Led by Alex and Jenna	1	8/15/2026	Plywood, Paint, Screws, Wood Glue	Home Depot
Grow light pulleys	Team	1	9/14/2026	Rope, Clip hangers, Pulleys	Shop in Kaitlyn's Family Home
Alternator and Electrical Box	Team, Led by Ethan and Carson	2	10/12/2026	Wood	Shop in Kaitlyn's Family Home

Part Images and Planning Needs



Fig 1. Bicycle Frame

- Budget Increase due to Fundraisers and Academic Donations
- Purchase bicycle frame from online store

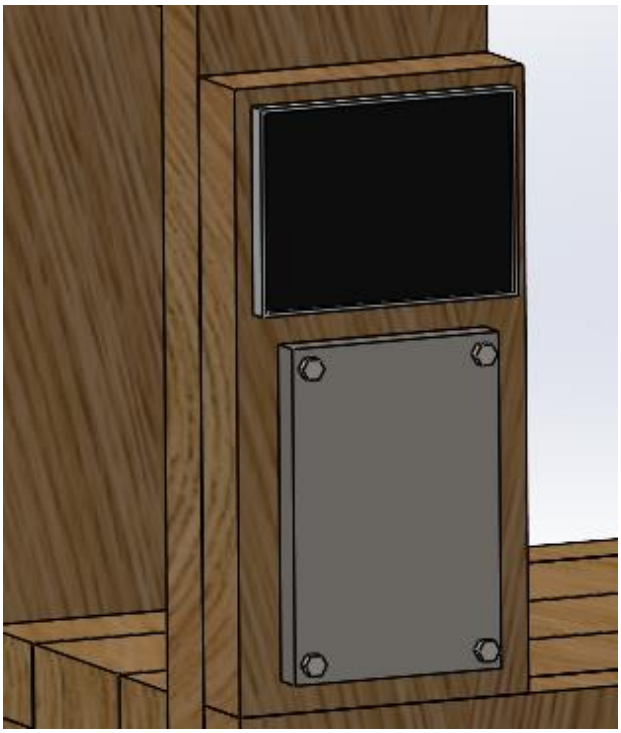


Fig 2. Electrical Box

- Finalize circuit
- Ensure alternator from Professor Pete is functional
- Determine alternator circuit requirements, or if DC motor is a better option

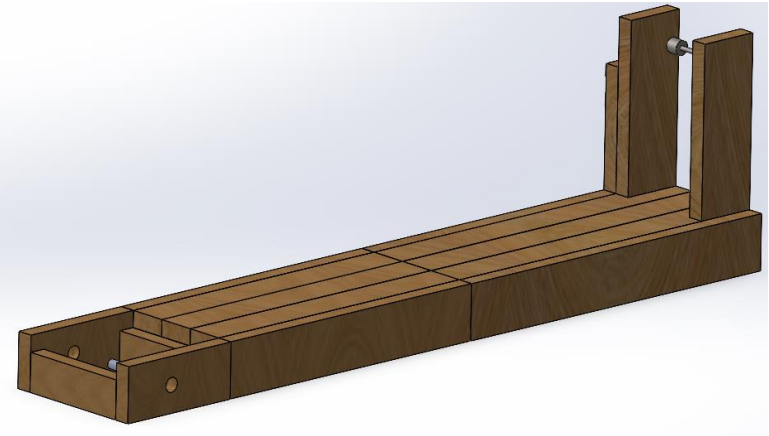


Fig 3. Frame Stand

- Due to new budget increase, new designs to be considered
- Plan to discuss new design

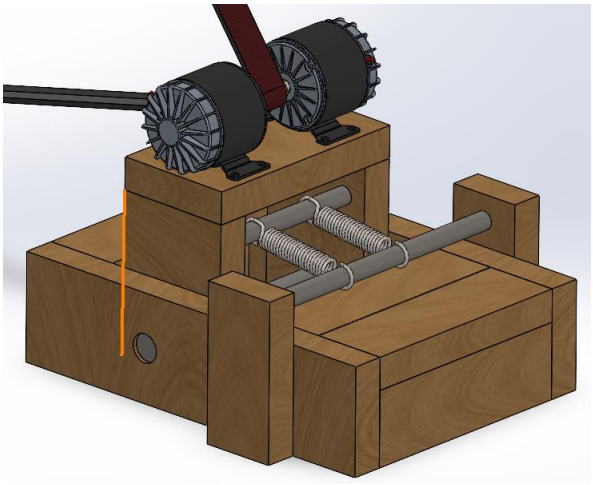


Fig 4. Spring System for Belt Power Generator

- Calculate spring constant requirements to inform purchasing
- Implement new frame stand considerations to current design

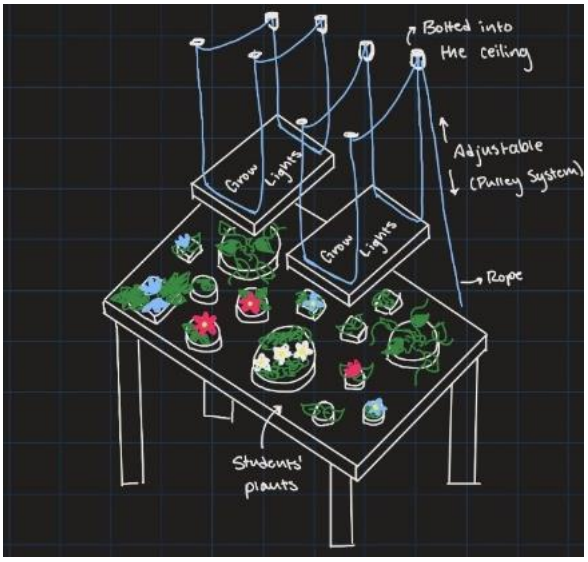


Fig 5. Grow Light Pulley System

- Calculate grow light load
- Set up grow light wiring with pulley system

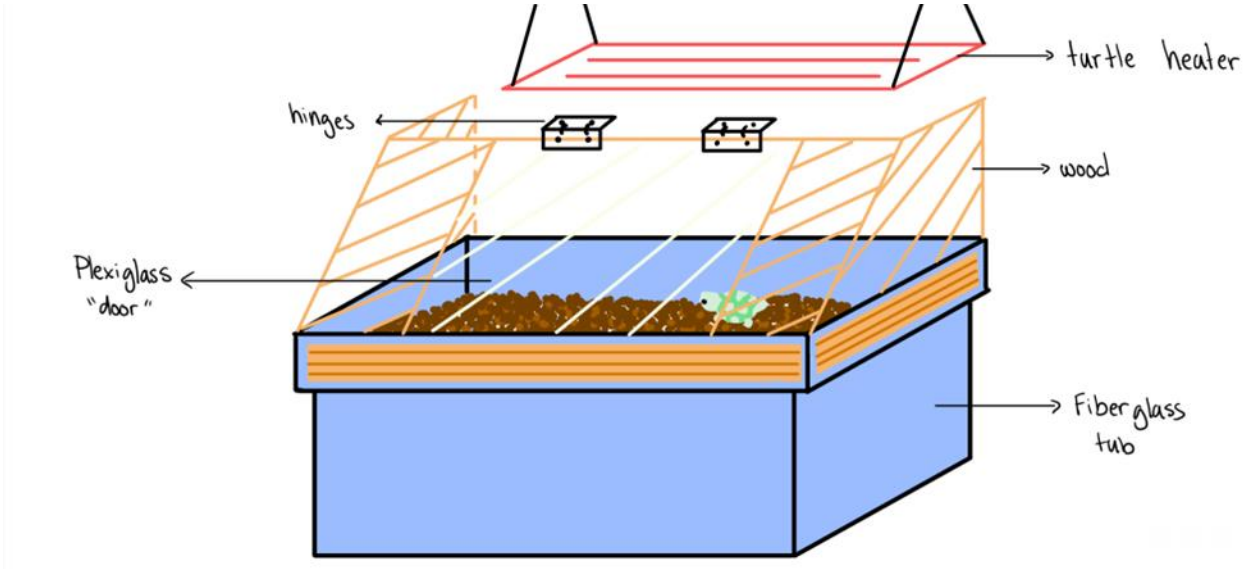


Fig 6. Sketch of Tortoise Habitat Enclosure

- Order materials
- Build habitat with the students

Action Items for 486C

1) Confirm donated materials for frame
2) Finalize dimensions with donated bike
3) Submit machine shop request before 7/27/2026, to give 8 weeks of time for manufacturing
4) Test belt vs. direct contact with prototype
5) Finalize generator circuit components and wiring

6) Calculate grow light load to inform pulley design
7) Order materials and set up PHS build date for tortoise enclosure

VI. Capstone II Deliverable Alterations

Proposal

With the final report of the project, the team is considering a final report binder to hand off to Ponderosa Highschool as well as Professor Carson Pete with the binder including all reports from this and the previous semester, circuit diagrams, CAD models, printouts and descriptions of all lesson plans (including slides), details of the improved battery and solar systems, and a subsection of PHS's documents given to the group last semester. This will ensure that if a future group of students want to continue working with the school and greenhouse or plan to do a similar project, these documents can serve as a guide/potential outline and can lead to future improvements.

VII. Conclusion

In this document, a draft for project management is laid out for the PHS sustainability capstone project. A reflection highlighting the success and room for improvement from ME476C is described with a focus on project management and team communication. A detailed Gantt chart along with the work breakdown schedule is provided to showcase the estimated schedule for next semester. The purchasing plan with a detailed BoM in the appendix shows how the team plans to get all items purchased and prepared for manufacturing. Lastly, the manufacturing plan describes how all items will be prepared. This document shows a draft of how the team plans to stay organized and execute all project goals in the final semester of the PHS sustainability capstone project.

VII. Appendix A: Bill of Materials

Category	Component	Model/SKU	Quantity	Unit $	Total $	Notes/description	Vendor
Solar	Solar screws for mounting solar	S-5 Hardware - specalized AL total quanity 250 pc	1	58.95	58.95	engineered mounting hardware for solar roof mounting	Metal Plus
Solar	sealant	Henry construction roof sealant	2	10.98	21.96	used for sealing roofing holes for solar	Home Depot
Outlets	screws	1lb construction screws 1.5"	1	9.58	9.58	used for mounting emt/boxes, etc	Donated
Outlets	wing nuts	100 pack wire nuts	1	17.98	17.98	use for wire connections inside	Donated
Outlets	EMT Backets/hangers	1 hole 1/2" emt straps - 25 pack	2	6.41	12.82	support EMT	Donated
Outlets	EMT Backets/hangers	1 pack of 1/2 hangers	2	4.81	9.62	support EMT	Donated
Outlets	EMT	half inch EMT 300 feet	30	7.51	225.3	new outlets and solar wire routing on roof	Home Depot
Outlets	EMT Connectors	1/2" connectors - 5 per pack	5	4.28	21.4	for indoor expansion of outlets	Donated
Outlets	EMT couplings	1/2" couplings - 5 per pack	5	3.96	19.8	for indoor expansion of outlets	Donated
Outlets	outlets	gfci outlets - pack	1	56.98	56.98	gfci outlets to protect new circuits	Donated
Outlets	outlets	outlets - normal	3	0.92	2.76	new outlets for indoors	Donated
Outlets	Breakers	20 A Square D - 1pole	2	6.98	13.96	New breakers 20 a	Donated

Outlets	Wire	12 gage thhn wire - white black and green	1	134	134	required for new wire indoors	Donated
Outlets	Wire	12 gage thhn wire - white black and green	1	134	134	required for new wire indoors	Donated
Outlets	Wire	12 gage thhn wire - white black and green	1	134	134	required for new wire indoors	Donated
Outlets	Wire	12 gage grounding pig tails for j boxes	1	6.61	6.61	grounding pig tails	Donated
Outlets	J-box cover	4" j box blank	2	1.34	2.68	for indoor j box	Donated
Outlets	4 inch j box	4" 2 gang j-boxes	6	8.99	53.94	for indoor new outlets or j box routes	Donated
Outlets	J-box cover	for switch and outlet	1	2.48	2.48	for door switch	Donated
Outlets	J-box cover	two gang cover for outlets	3	2.48	7.44	for new outlets	Donated
Solar	Battery	DC400-6	4	823.13	3292.52	Fullriver Battery DC400-6 Sealed AGM Deep Cycle 6 Volts 415Ah	NAZ Solar
Solar	Cables	UL-2012-BK	3	32.34	97.02	Battery Interconnect Cable #2/0 AWG x 12" MTW Rated UL Listed in Black	NAZ Solar
Solar	Cables	UL-2060-PR	1	169.65	169.65	Inverter Cables Red/Black Pair, #2/0 AWG, 5 Foot	NAZ Solar

						UL Listed MTW	
Solar	MC 4 Connectors - male	18-32.0017P0001-UR	25	2.38	59.5	Male MC4 Connector Contact with Housing	NAZ Solar
Solar	MC 4 Connectors - female	18-32.0016P0001-UR	25	2.38	59.5	Female MC4 Connector Housing and Contact	NAZ Solar
Solar	MC4 Parallel Multibranch Adapters	SLCBL-46	4	12.82	51.28	Solarland SLCBL-46 MC4 Parallel Branch Connectors Set	NAZ Solar
Solar	S-5! CorruBracket	CorruBracket	18	9.36	168.48	mounting on corrugated roof	NAZ Solar
Solar	Solar rails	91290 - 2.3-Rail-BF-93-2-Kit	1	216.97	216.97	Tamarack Solar 91290 2.3 Rail 93-inch Black 2 Rail Kit	NAZ Solar
Solar	Solar rails	Tamarack Solar 90514 Flush Mount Kits 2.3 Black Rail 93-inch – 4 Rail Kit	1	395.63	395.63	Tamarack Solar 90514 Flush Mount Kits 2.3 Black Rail 93-inch – 4 Rail Kit	NAZ Solar
Solar	Solar rails	88412	1	21.64	21.64	Tamarack Solar 88412 2.3 Push-In Bonding Rail Splice	NAZ Solar
Prototype	Bicycle		1	0	0	1 non shifting bike donated to use to create the human powered	Donated

						energy generation stationary bike	
Prototype	Alternator		1	0	0	Converts mechanical energy into electrical energy	Donated
Prototype	Drive Belt		1	0	0	72.30 in Serpentine Belt	Donated
Prototype	Concord BMX Freestyle Bike Pegs	# 4849	1	10.89	10.89	Alloy bike pegs used to fix bike to the frame stand.	Walmart
Prototype	7/16" Bolts	# 2451	1	11.15	11.15	Bolts added to fix alternator	Home Depot
Greenhouse	Pulley System	M25-4MM	1	23.03	23.03	M25 Stainless Steel Pulley System Kit	Walmart
Greenhouse	Plexiglass		1	0	0	Optix 18in x 24in x 0.093 (3/32) in clear acrylic sheet	Donated
Greenhouse	Wood	# 458503	4	13.62	54.48	12x48 inch ponderosa pine	Home Depot
Greenhouse	Screws	# 442472	1	2.66	6.97	#8 1in screws	Home Depot
Greenhouse	Wood Glue	# 1003827526	2	3.98	7.96	4fl oz wood glue	Home Depot
Greenhouse	Paint	# 705004	4	6.98	27.92	8 ounce cans(red, blue, yellow, black)	Home Depot
Greenhouse	Paint	# 705004	1	18.48	18.48	1 quart white	Home Depot
Greenhouse	LED Light strands		2	129.79	259.58	Waterproof LED Neon Rope Light Milti-Color	The DIY Outlet

						App RF Remote	
General	Kill A Watt Electrical Monitor	93519/4400	1	29.99	29.99	Device that monitors and records electrical expenses by the hour, day, week, month, and year.	Harbor Freight
General	Multimeter	# 59410	1	24.99	24.99	Measures AC/DC voltage to 600 volts, DC current, resistance, diode, capacitance, audible continuity, and battery test.	Harbor Freight
Bicycle	Bicycles		2	0	0	2 mountain bike with gears donated to use to create the human powered energy generation stationary bike	Donated
Bicycle	Alternator		2	0	0	Converts mechanical energy into electrical energy	Donated
Bicycle	Wood	2 inches by 4 inches	12	3.85	46.2	To build a wooden frame to fix the back tire of the bicycles and	Home Depot

						hold them in place	
Bicycle	Plywood Sheets	# 386081	2	11.93	35.79	Flat playwood surface to be place under the frame and secured	Home Depot
Bicycle	Prime-Line carriage nuts and bolts	GD 52103	10	3.3	30.3	1/4 inch-20 teeth per inch (TPI) measuring 1/4 inch diameter x 1-7/8 inch length.	Home Depot
Bicycle	Metal Rod	17R.875-72	1	57.07	57.07	Carbon Steed Rod 7/8 inch 6ft long, cold rolled	Grainger
Bicycle	LED Display		2	50.99	101.98	Waveshare 7.5 inch display screen	Amazon
Bicycle	Bike Frame		1	$55.99	$55.99	FDW Magnetic Indoor Bicycle Brike Trainer Exercise Stand	Walmart
		TOTAL PARTS:	209	**TOTAL VALUE:**	6281.22		